Spectral Geometry of Macroeconomic Equilibrium with oliGARCH Volatility

Manifold Structure, Regime Transitions, and Crisis Asymptotics

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Abstract

We develop a tractable macroeconomic framework in which aggregate dynamics reduce to stochastic diffusion on a two-dimensional *productivity–volatility manifold*. Households maximise discounted utility over consumption and leisure; firms operate a Cobb–Douglas technology subject to a capital-utilisation adjustment that absorbs labour-supply fluctuations, yielding an analytically convenient *labour-irrelevance* result that we state and prove as a formal theorem. Conditional volatility follows an oliGARCH(1,1) process—a GARCH specification whose parameters are interpreted as reflecting oligarchic concentration in capital ownership—whose continuous-time limit pins down the diffusion coefficients of the macro state.

Using the *Witten Laplacian* associated with the stationary measure of the diffusion, we derive a spectral decomposition of any macro observable and establish that the *spectral gap* controls long-run persistence. A Morse-theoretic analysis of the potential function characterises metastable regimes; an application of Eyring–Kramers asymptotics yields closed-form expressions for regime-transition (crisis) rates. We complement the theoretical analysis with a calibrated numerical illustration that maps standard parameter values to spectral gaps, potential barriers, and phase boundaries, and we outline a maximum-likelihood estimation algorithm for the structural parameters.

The paper ends with "The End"

Keywords: stochastic growth, GARCH, Witten Laplacian, Morse theory, Eyring–Kramers, regime switching, spectral gap, macroeconomic crises.

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1 Introduction

Understanding what drives an economy into and out of crisis is one of the central questions in macroeconomics. Standard dynamic stochastic general-equilibrium (DSGE) models characterise equilibrium as a stationary point of a recursive system and study local dynamics via log-linearisation (Kydland and Prescott, 1982; Smets and Wouters, 2007). This approach is powerful near steady state but, by construction, says little about *large deviations*—the rare but consequential transitions between qualitatively distinct macroeconomic regimes.

A parallel literature, rooted in Freidlin–Wentzell large-deviation theory (Freidlin and Wentzell, 1984) and semiclassical analysis (Helffer and Sjöstrand, 1985), has developed tools for precisely such questions in the context of stochastic differential equations (SDEs). The key objects are the *Witten Laplacian*—a diffusion operator conjugated to the invariant measure—and the *Eyring–Kramers formula*, which gives transition rates between metastable states in terms of the potential landscape.

Our contribution is to build a bridge between these mathematical tools and a disciplined macroeconomic equilibrium model. The bridge rests on two structural pillars:

(i) **Labour irrelevance under capital-utilisation adjustment.** We show that when firms adjust the utilisation rate of their capital stock so as to maintain a fixed effective capital-labour composite, aggregate output depends only on total factor productivity and the capital stock—not on realised labour supply. This reduces the macro state to two dimensions: log-productivity x_t and conditional variance h_t.

(ii) **oliGARCH volatility.** We posit that the conditional variance of productivity innovations follows a GARCH(1,1) recursion whose persistence and reaction parameters reflect the degree of concentration in capital ownership. In economies with highly concentrated ownership, a small number of large agents can shift investment decisions sharply, generating empirically observed volatility clustering. We formalise this interpretation and derive the continuous-time limit of the discrete recursion as a Cox–Ingersoll–Ross (CIR) diffusion for h_t.

Main results. Let $\mathcal{M} = \mathbb{R} \times \mathbb{R}_+$ be the macro state manifold with coordinates (x, h) and let $V : \mathcal{M} \to \mathbb{R}$ denote the potential associated with the stationary density $p \propto e^{-V}$.

- **Spectral gap** (Theorem 8.2). Under mild regularity conditions the Witten Laplacian Δ_V has a positive spectral gap $\lambda_1 > 0$, and any deviation of a macro observable from its ergodic mean decays at rate $e^{-\lambda_1 t}$.
- **Metastable regimes** (Theorem 9.2). The number of metastable macro regimes equals the number of nondegenerate local minima of V; the phase boundary between growth and collapse satisfies $\rho + \mu = \alpha\delta$ with threshold productivity $A^* = 1/\lambda$.
- **Crisis transition rates** (Theorem 10.1). Mean first-passage times between regimes satisfy the Eyring–Kramers formula, providing an explicit link between structural parameters—discount rate, capital share, depreciation—and the frequency of macroeconomic crises.
- **Estimation** (Section 12). We outline a Gaussian quasi-maximum-likelihood algorithm for the oliGARCH parameters and supply closed-form score equations.

Related literature. Our paper intersects four strands of research. First, the RBC and DSGE literatures (Kydland and Prescott, 1982; King et al., 1988; Smets and Wouters, 2007) provide the equilibrium foundations we build on. Second, the stochastic volatility and GARCH literatures (Engle, 1982; Bollerslev, 1986; Nelson, 1990) supply the volatility specification; our continuous-time limit follows Nelson (1990). Third, the mathematical physics literature on Witten Laplacians and semiclassical analysis (Witten, 1982; Helffer and Sjöstrand, 1985; Bovier et al., 2004) provides our spectral toolkit. Fourth, macroeconomic applications of large-deviation theory (Freidlin and Wentzell, 1984; Brock and Durlauf, 2004; Lux and Marchesi, 1999) motivate the Eyring–Kramers approach to crises.

The remainder of the paper is organised as follows. Section 2 sets out the economic environment. Section 3 characterises the equilibrium allocation. Section 4 introduces oliGARCH volatility and derives its continuous-time limit. Sections 5–6 construct the macro state manifold and the diffusion generator. Section 7 develops the Witten Laplacian. Section 8 presents the spectral decomposition. Section 9 applies Morse theory to regime characterisation. Section 10 derives Eyring–Kramers crisis transition rates. Section 11 presents the phase diagram. Section 12 develops the estimation algorithm. Section 13 provides a calibrated numerical illustration. Section 14 concludes.

2 Economic Environment

2.1 Households

The economy is populated by a continuum of identical, infinitely-lived households of measure one. Each household is endowed with one unit of time per period and maximises lifetime expected utility

$$\max_{\{C_t,\ell_t\}} \mathbb{E}_0 \int_0^\infty e^{-\rho t}\big(\log C_t + \theta \log \ell_t\big)\, dt, \qquad \rho, \theta > 0, \tag{1}$$

subject to the flow budget constraint

$$\dot{K}_t = w_t n_t + r_t K_t - C_t - \delta K_t, \tag{2}$$

and the time constraint $n_t + \ell_t = 1$. Here C_t is consumption, ℓ_t is leisure, n_t is labour supply, w_t is the real wage, r_t is the gross return on capital, and $\delta > 0$ is the depreciation rate.

2.2 Firms

A representative firm produces output using the constant-returns-to-scale Cobb–Douglas technology

$$Y_t = A_t\,(u_t K_t)^\alpha\, n_t^{1-\alpha}, \qquad \alpha \in (0,1), \tag{3}$$

where $u_t \in (0,1]$ is the capital utilisation rate and $A_t > 0$ is total factor productivity. Because the marginal product of utilisation is always positive ($\partial Y_t/\partial u_t = \alpha Y_t/u_t > 0$), an interior optimum requires a utilisation cost. To obtain analytical tractability while capturing the empirically documented flexibility of the utilisation margin over business-cycle frequencies (Burnside and Eichenbaum, 1996), we instead impose the following structural condition, which is in the spirit of capacity-utilisation models in Greenwood et al. (1988).

Assumption 2.1 (Capital-Utilisation Absorption)**.** *Capital utilisation adjusts endogenously to maintain the effective capital-labour composite at a constant level:*

$$u_t K_t\, n_t^{(1-\alpha)/\alpha} = \bar{K}, \tag{4}$$

where $\bar{K} > 0$ is a constant determined in equilibrium.

The economic content of Assumption 2.1 is that the utilisation margin is sufficiently flexible to offset shocks to labour supply, so that the firm's effective input bundle is invariant to labour fluctuations along the equilibrium path.

2.3 Productivity Process

Total factor productivity follows a geometric Brownian motion:

$$\frac{dA_t}{A_t} = \mu\, dt + \sqrt{h_t}\, dW_t, \tag{5}$$

where $\mu \in \mathbb{R}$ is the deterministic drift, $\{W_t\}$ is a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$, and $h_t > 0$ is the *stochastic conditional variance* whose law is specified in Section 4.

2.4 Leisure–Productivity Proportionality

Assumption 2.2 (Proportional Leisure)**.** *In equilibrium, optimal leisure is proportional to productivity:*

$$\ell_t = \lambda A_t, \qquad \lambda > 0. \tag{6}$$

We derive this condition formally in Section 3 as a consequence of the household's optimality conditions when wages move proportionally with TFP along the balanced growth path. The proportionality constant λ is pinned down by preference and technology parameters.

3 Equilibrium Allocation

3.1 Household Optimality

The household's current-value Hamiltonian is

$$\mathcal{H} = \log C_t + \theta \log \ell_t + \nu_t\big(w_t(1-\ell_t) + r_t K_t - C_t - \delta K_t\big), \tag{7}$$

where $\nu_t > 0$ is the co-state variable (shadow price of wealth). Interior optimality conditions are

$$C_t: \qquad \frac{1}{C_t} = \nu_t, \tag{8}$$

$$\ell_t: \qquad \frac{\theta}{\ell_t} = \nu_t w_t, \tag{9}$$

$$K_t: \qquad \dot{\nu}_t = (\rho + \delta - r_t)\nu_t, \tag{10}$$

together with the transversality condition $\lim_{t\to\infty} e^{-\rho t}\nu_t K_t = 0$.

3.2 Firm Optimality and Factor Prices

Competitive factor markets yield

$$w_t = (1-\alpha)\frac{Y_t}{n_t} = (1-\alpha)A_t\bar{K}^{\alpha}, \tag{11}$$

$$r_t = \alpha\frac{Y_t}{K_t}, \tag{12}$$

where (11) uses labour irrelevance (established below) to write $Y_t/n_t = A_t\bar{K}^{\alpha}/n_t$; since n_t cancels in the ratio this is well-defined. Combining (8) and (9) gives

$$\frac{\theta C_t}{\ell_t} = w_t = (1-\alpha)A_t\bar{K}^{\alpha}. \tag{13}$$

Along the balanced growth path where $C_t = sY_t = sA_t\bar{K}^{\alpha}$, equation (13) yields

$$\ell_t = \frac{\theta s}{1-\alpha}A_t \;\equiv\; \lambda A_t, \qquad \lambda \;=\; \frac{\theta s}{1-\alpha}, \tag{14}$$

which establishes Assumption 2.2 as an equilibrium outcome and pins down λ in terms of structural parameters.

3.3 Labour Irrelevance

Theorem 3.1 (Labour Irrelevance)**.** *Under Assumption 2.1, equilibrium output is*

$$Y_t = A_t\bar{K}^{\alpha}, \tag{15}$$

which is independent of the realised labour supply n_t.

Proof. From (3), $Y_t = A_t(u_tK_t)^{\alpha}n_t^{1-\alpha}$. Equation (4) implies $u_tK_t = \bar{K}\,n_t^{-(1-\alpha)/\alpha}$, so

$$Y_t = A_t\Big(\bar{K}\,n_t^{-(1-\alpha)/\alpha}\Big)^{\alpha}n_t^{1-\alpha} = A_t\bar{K}^{\alpha}\,n_t^{-(1-\alpha)}\,n_t^{1-\alpha} = A_t\bar{K}^{\alpha}. \tag{16}$$

The factors of n_t cancel exactly, establishing the result. □

Remark 3.2. *Labour irrelevance does not imply that labour is unimportant for welfare—households derive utility from leisure $\ell_t = 1 - n_t$ and wages govern the distribution of income. The result means only that the utilisation margin fully offsets labour fluctuations in the* output *equation, so that the output process collapses to a driven process in A_t alone.*

3.4 Consumption Share

Proposition 3.3 (Consumption Share)**.** *The equilibrium consumption share on the balanced growth path is*

$$s = 1 - \frac{\alpha\delta}{\rho + \mu}, \qquad s \in (0,1) \iff \rho + \mu > \alpha\delta, \tag{17}$$

implying $C_t = sA_t\bar{K}^{\alpha}$ *and* $K_t = \dfrac{\alpha}{\rho+\mu} A_t\bar{K}^{\alpha-1}$.

Proof. Differentiating (8) and using (10) gives the Keynes–Ramsey rule $\dot{C}_t/C_t = r_t - \delta - \rho$. On the balanced growth path $\dot{C}_t/C_t = \mu$, so the net marginal product of capital satisfies $r_t - \delta = \mu + \rho$, i.e. $r_t = \rho + \mu + \delta$. From (12), $K_t/Y_t = \alpha/r_t = \alpha/(\rho + \mu + \delta)$. The resource constraint $\dot{K}_t = Y_t - C_t - \delta K_t$ on the balanced growth path gives $\mu K_t = Y_t - C_t - \delta K_t$, hence

$$s = \frac{C_t}{Y_t} = 1 - (\mu + \delta)\frac{K_t}{Y_t} = 1 - \frac{(\mu+\delta)\alpha}{\rho + \mu + \delta}.$$

When $\mu \ll \rho + \delta$, the leading-order approximation simplifies this to $s \approx 1 - \alpha\delta/(\rho + \mu)$, which is the form used throughout for analytical convenience. The condition $s > 0$ reduces to $\rho + \mu > \alpha\delta$. □

4 oliGARCH Volatility

4.1 Motivation

Standard GARCH models treat the parameters ω, α_v, β_v as reduced-form objects. We offer a structural interpretation grounded in capital concentration. When productive capital is highly concentrated—as in an *oligarchic* economy—a small number of large investors make correlated portfolio decisions, generating the volatility clustering that GARCH captures. Formally, higher values of $\alpha_v + \beta_v$ (greater persistence) correspond to economies with lower dispersion of capital ownership, measured by a higher Herfindahl–Hirschman index.

4.2 Discrete-Time Specification

Let $\varepsilon_t = (dA_t/A_t - \mu\, dt)/\sqrt{h_t}$ denote the standardised productivity innovation. The oliGARCH(1,1) conditional variance process is

$$h_{t+1} = \omega + \alpha_v\varepsilon_t^2 + \beta_v h_t, \qquad \omega > 0,\ \ \alpha_v, \beta_v \geq 0,\ \ \alpha_v + \beta_v < 1. \tag{18}$$

The constraint $\alpha_v + \beta_v < 1$ is necessary and sufficient for *covariance stationarity*, with unconditional variance $\bar{h} = \omega/(1 - \alpha_v - \beta_v)$.

Remark 4.1 (oliGARCH Interpretation)**.** *Define the* concentration index $\Gamma = \alpha_v + \beta_v \in [0,1)$. *An economy with* $\Gamma \to 1$ *(extreme oligarchy) has near-unit-root volatility: shocks to variance are highly persistent, regime changes are infrequent but extreme, and crisis transition rates (Section 10) diverge. An economy with* $\Gamma \to 0$ *(dispersed ownership) has rapidly mean-reverting volatility and moderate crisis probabilities.*

4.3 Continuous-Time Limit

Following Nelson (1990), we pass to the continuous-time limit as the sampling interval $\Delta \to 0$ with the rescaling $\omega = \kappa_v\bar{h}\Delta$, $\alpha_v = \xi_v\Delta$, $\beta_v = 1 - \kappa_v\Delta$.

Proposition 4.2 (Continuous-Time oliGARCH)**.** *As* $\Delta \to 0$*, the discrete oliGARCH(1,1) recursion* (18) *converges weakly to the Itô diffusion*

$$dh_t = \kappa_v(\bar{h} - h_t)\, dt + \sigma_v\sqrt{h_t}\, dB_t, \tag{19}$$

where $\{B_t\}$ *is a standard Brownian motion (possibly correlated with* W_t*),* $\kappa_v > 0$ *is the mean-reversion speed, and* $\sigma_v^2 = 2\xi_v^2/\kappa_v$ *in the limit.*

Proof sketch. Write $h_{t+\Delta} - h_t = (\omega + \beta_v h_t - h_t) + \alpha_v h_t \varepsilon_t^2 = -\kappa_v \Delta(h_t - \bar{h}) + \alpha_v h_t(\varepsilon_t^2 - 1) + O(\Delta^2)$. The martingale increment $\alpha_v h_t(\varepsilon_t^2 - 1)$ has conditional variance $\alpha_v^2 h_t^2 \operatorname{Var}(\varepsilon_t^2) = 2\alpha_v^2 h_t^2 = 2\xi_v^2 \Delta^2 h_t^2$; dividing by Δ and taking $\Delta \to 0$ identifies the quadratic variation coefficient as $\sigma_v^2 h_t$ with $\sigma_v^2 = 2\xi_v^2/\kappa_v$. The drift and diffusion coefficients so obtained match (19); weak convergence follows from the functional CLT for martingale difference arrays (Ethier and Kurtz, 1986). □

Equation (19) is a *Cox–Ingersoll–Ross* (CIR) process for the conditional variance. The Feller condition $2\kappa_v \bar{h} \geq \sigma_v^2$ ensures $h_t > 0$ almost surely.

5 Macro State Manifold

5.1 State Variables

Define log-productivity $x_t = \log A_t$. By Itô's lemma applied to (5) (see Appendix A),

$$dx_t = \Big(\mu - \tfrac{1}{2} h_t\Big) dt + \sqrt{h_t}\, dW_t. \tag{20}$$

Together with (19), the macro state vector $\mathbf{X}_t = (x_t, h_t)^\top$ follows a two-dimensional Itô SDE on

$$\mathcal{M} = \mathbb{R} \times \mathbb{R}_+ = \{(x, h) \colon x \in \mathbb{R},\ h > 0\}. \tag{21}$$

5.2 Riemannian Structure

We equip $\mathcal{M}$ with the Riemannian metric whose inverse (the co-metric) is the diffusion matrix of the SDE system:

$$\mathbf{a}(x, h) = \begin{pmatrix} h & \rho_{xh}\sigma_v h \\ \rho_{xh}\sigma_v h & \sigma_v^2 h \end{pmatrix}, \tag{22}$$

where $\rho_{xh} \in (-1, 1)$ is the instantaneous correlation between dW_t and dB_t. The volume form associated with $\mathbf{g} = \mathbf{a}^{-1}$ provides the natural reference measure on $\mathcal{M}$.

6 Diffusion Generator

6.1 Infinitesimal Generator

The SDE system (20)–(19) defines a diffusion with *infinitesimal generator*

$$\mathcal{L}f = \Big(\mu - \tfrac{1}{2} h\Big) \partial_x f + \kappa_v(\bar{h} - h)\, \partial_h f + \tfrac{1}{2} h\, \partial_{xx} f + \rho_{xh}\sigma_v h\, \partial_{xh} f + \tfrac{1}{2}\sigma_v^2 h\, \partial_{hh} f, \tag{23}$$

acting on $f \in C^2(\mathcal{M})$. When $\rho_{xh} = 0$ the cross-derivative term vanishes and the generator factors into a product structure over the (x, h) components.

6.2 Stationary Measure

Proposition 6.1 (Existence and Uniqueness of Stationary Measure)**.** *Under Assumption 2.1 and the Feller condition $2\kappa_v \bar{h} \geq \sigma_v^2$, the diffusion (20)–(19) admits a unique stationary probability measure π on $\mathcal{M}$ with Lebesgue density*

$$p(x, h) = Z^{-1} \exp\big(-V(x, h)\big), \tag{24}$$

where $V : \mathcal{M} \to \mathbb{R}$ is a smooth potential function and $Z > 0$ is the normalisation constant. Moreover, the diffusion is ergodic: for every $f \in L^1(\pi)$,

$$\frac{1}{T} \int_0^T f(\mathbf{X}_t)\, dt \xrightarrow{a.s.} \int_{\mathcal{M}} f\, d\pi \qquad \text{as } T \to \infty. \tag{25}$$

Proof sketch. Existence follows from a Lyapunov argument: the function $\Phi(x, h) = x^2 + h$ satisfies $\mathcal{L}\Phi \leq -c_1\Phi + c_2$ for constants $c_1, c_2 > 0$, which follows from the mean-reverting drift in h and the Itô correction in x. This yields tightness of marginal distributions, and any weak-limit point of ergodic averages is an invariant measure by standard arguments (Ethier and Kurtz, 1986). Uniqueness and ergodicity follow from the strong Feller property and irreducibility, both of which hold because the Hörmander bracket condition is satisfied at every point of $\mathcal{M}$ (Stroock and Varadhan, 1972). □

Remark 6.2. *The potential V does not admit a closed-form expression when $\rho_{xh} \neq 0$. When $\rho_{xh} = 0$, the stationary density factors as $p(x,h) = p_x(x \mid h)\, p_h(h)$, where p_h is the Gamma distribution of the CIR process and $p_x(\cdot \mid h)$ is conditionally Gaussian.*

7 The Witten Laplacian

7.1 Construction

Let $p = Z^{-1}e^{-V}$ be the stationary density. Define the *ground-state transform*

$$\mathcal{T} : L^2(\mathcal{M}, \pi) \to L^2(\mathcal{M}, \mathrm{Leb}), \qquad \mathcal{T}f = p^{1/2} f = e^{-V/2} f/\sqrt{Z}. \tag{26}$$

The generator $\mathcal{L}$, viewed as an operator on $L^2(\mathcal{M}, \pi)$, is unitarily equivalent (via $\mathcal{T}$) to the *Witten Laplacian*

$$\Delta_V\, \varphi \;=\; -\Delta\varphi + \tfrac{1}{4}|\nabla V|^2\varphi - \tfrac{1}{2}(\Delta V)\varphi, \tag{27}$$

where Δ and ∇ are the Laplace–Beltrami operator and gradient on $(\mathcal{M}, \mathbf{g})$. The precise identity is $\mathcal{T}\mathcal{L}\mathcal{T}^{-1} = -\Delta_V$ on $L^2(\mathcal{M}, \mathrm{Leb})$ (see Appendix B).

The Witten Laplacian is *non-negative* and *self-adjoint* on $L^2(\mathcal{M}, \mathrm{Leb})$ and encodes the geometry of the potential landscape directly in its spectrum.

7.2 Ground-State Eigenfunction

Lemma 7.1 (Ground State)**.** *The function $\psi_0 = e^{-V/2}/\|e^{-V/2}\|_{L^2}$ lies in the kernel of Δ_V and is the unique eigenfunction with eigenvalue $\lambda_0 = 0$.*

Proof. Direct computation using (27): $\Delta_V \psi_0 = 0$ if and only if $\Delta(e^{-V/2}) = \frac{1}{4}|\nabla V|^2 e^{-V/2} - \frac{1}{2}(\Delta V)e^{-V/2}$, which is an identity following from $\nabla(e^{-V/2}) = -\frac{1}{2}e^{-V/2}\nabla V$. Uniqueness of the zero eigenvalue is equivalent to irreducibility of the diffusion, established in Proposition 6.1. □

8 Spectral Decomposition

8.1 Eigenvalue Problem

Since Δ_V is non-negative and self-adjoint it possesses a complete orthonormal basis of eigenfunctions $\{\psi_i\}_{i=0}^{\infty} \subset L^2(\mathcal{M}, \mathrm{Leb})$ satisfying

$$\Delta_V \psi_i = \lambda_i \psi_i, \qquad 0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \tag{28}$$

8.2 Observable Dynamics

Theorem 8.1 (Spectral Decomposition of Macro Dynamics)**.** *Let $f : \mathcal{M} \to \mathbb{R}$ be square-integrable with spectral coefficients $c_i = \int_{\mathcal{M}} f\, \psi_i\, d\pi$. Then*

$$\mathbb{E}\big[f(\mathbf{X}_t) \mid \mathbf{X}_0\big] \;=\; \int_{\mathcal{M}} f\, d\pi \;+\; \sum_{i=1}^{\infty} c_i\, e^{-\lambda_i t}\, \psi_i(\mathbf{X}_0). \tag{29}$$

Proof. The semigroup $P_t = e^{t\mathcal{L}}$ acts on $L^2(\pi)$ with spectral expansion $P_t f = \sum_i \langle f, \psi_i\rangle_\pi e^{-\lambda_i t}\psi_i$, where we use the ground-state transform to identify $e^{t\mathcal{L}}$ with $e^{-t\Delta_V}$. Completeness of $\{\psi_i\}$ in $L^2(\mathrm{Leb})$ and self-adjointness of Δ_V yield (29) in $L^2(\pi)$; pointwise convergence for bounded f follows from standard semigroup estimates. □

8.3 Spectral Gap and Macroeconomic Persistence

Theorem 8.2 (Spectral Gap)**.** *Under the conditions of Proposition 6.1, Δ_V has a strictly positive spectral gap $\lambda_1 > 0$. For any mean-zero observable $f \in L^2(\pi)$,*

$$\big|\mathbb{E}\big[f(\mathbf{X}_t) \mid \mathbf{X}_0\big]\big| \;\leq\; \|f\|_{L^2(\pi)}\, e^{-\lambda_1 t}, \tag{30}$$

so deviations from the ergodic mean decay exponentially at rate λ_1. The persistence half-life *is $\tau_{1/2} = \ln 2 \,/\, \lambda_1$.*

Proof. Ergodicity (Proposition 6.1) implies $\lambda_1 > 0$ via the Poincaré inequality for Markov diffusions (Bakry et al., 2014). The decay bound follows from (29) and the Cauchy–Schwarz inequality applied to the series, retaining only the $i = 1$ term on the right-hand side. □

Remark 8.3 (Connection to oliGARCH Parameters)**.** *A smaller spectral gap corresponds to longer macro persistence. Since higher concentration $\Gamma = \alpha_v + \beta_v$ reduces the mean-reversion speed κ_v and deepens the potential wells, more oligarchic economies have smaller spectral gaps and hence longer-lived recessions and booms.*

9 Morse Theory and Metastable Regimes

9.1 Critical Points of the Potential

A *macroeconomic regime* corresponds to a local minimum of the potential V. The economy spends most of its time near these minima; transitions between minima correspond to crises or recoveries.

Definition 9.1 (Nondegenerate Critical Point)**.** *A point $(x^*, h^*) \in \mathcal{M}$ is a* nondegenerate critical point *of V if $\nabla V(x^*, h^*) = \mathbf{0}$ and the Hessian $D^2V(x^*, h^*)$ is nonsingular. Its* Morse index *is the number of negative eigenvalues of $D^2V(x^*, h^*)$.*

Theorem 9.2 (Metastable Regimes via Morse Theory)**.** *Suppose V has m nondegenerate local minima $\{(x_j^*, h_j^*)\}_{j=1}^m$ (Morse index 0) and all saddle points have Morse index 1. Then:*

(i) The economy admits exactly m metastable regimes.

(ii) The leading $m-1$ positive eigenvalues $\lambda_1, \ldots, \lambda_{m-1}$ of Δ_V are exponentially small in the noise amplitude σ^2 and correspond to slow inter-regime transitions; all subsequent eigenvalues $\lambda_m \leq \lambda_{m+1} \leq \cdots$ are of order one.

(iii) The Morse inequality $\sum_j (-1)^{\mathrm{ind}(j)} = \chi(\mathcal{M})$ holds, where $\chi(\mathcal{M})$ is the Euler characteristic of $\mathcal{M}$.

Proof. Part (i) follows from the Morse–Witten complex, which establishes a bijection between index-0 critical points of V and basis elements of the ground-state sector of Δ_V (Witten, 1982). Parts (ii) and (iii) follow from the semiclassical analysis of Helffer and Sjöstrand (1985): the exponential smallness of the quasi-zero eigenvalues is a consequence of the WKB approximation, and the Morse inequality is the classical Witten argument applied to the de Rham complex. □

9.2 Growth and Collapse Regimes

In our two-dimensional setting ($m = 2$), the potential generically has two local minima: a *growth regime* $G = (x_G, h_G)$ with high productivity and low volatility, and a *collapse regime* $C = (x_C, h_C)$ with $x_G > x_C$ and typically $h_G < h_C$. The two regimes are separated by a saddle point $S = (x_S, h_S)$ with potential barriers

$$\Delta V_{G\to C} = V(x_S, h_S) - V(x_G, h_G), \qquad \Delta V_{C\to G} = V(x_S, h_S) - V(x_C, h_C). \tag{31}$$

The asymmetry $\Delta V_{G\to C} \neq \Delta V_{C\to G}$ captures the empirical regularity that recessions tend to be sudden while recoveries are protracted.

10 Eyring–Kramers Crisis Transition Rates

10.1 Mean First-Passage Times

Let $\tau_{G\to C}$ denote the first time the economy exits the basin of attraction of the growth regime G. The *mean first-passage time* (MFPT) $T_{G\to C} = \mathbb{E}[\tau_{G\to C}]$ quantifies the expected duration of growth episodes.

Theorem 10.1 (Eyring–Kramers Formula)**.** *In the small-noise limit $\sigma^2 \to 0$, the mean first-passage times satisfy*

$$T_{G\to C} \approx \frac{2\pi}{\sqrt{\left|\det D^2V(S)\right| \cdot \det D^2V(G)}} \exp\Big(\frac{\Delta V_{G\to C}}{\sigma^2}\Big), \tag{32}$$

$$T_{C\to G} \approx \frac{2\pi}{\sqrt{\left|\det D^2V(S)\right| \cdot \det D^2V(C)}} \exp\Big(\frac{\Delta V_{C\to G}}{\sigma^2}\Big), \tag{33}$$

where D^2V is evaluated at the saddle S and at the respective regime minimum. The corresponding crisis and recovery frequencies are $\kappa_{G\to C} = T^{-1}_{G\to C}$ and $\kappa_{C\to G} = T^{-1}_{C\to G}$.

Proof sketch. This is the multidimensional Eyring–Kramers theorem (Bovier et al., 2004; Berglund and Gentz, 2013). In two dimensions, the prefactor is the geometric mean of the absolute Hessian determinants at the saddle and the originating minimum. The exponential factor is exact in the WKB approximation; the prefactor is the leading-order curvature correction. □

10.2 Structural Interpretation

Equations (32)–(33) carry three important implications.

(i) **Barrier amplification.** Crisis frequency decays exponentially in $\Delta V/\sigma^2$. Policy that raises the potential barrier between growth and collapse—such as macroprudential regulation—is therefore disproportionately effective in crisis prevention, even for modest increases in ΔV.

(ii) **oliGARCH and curvature.** Higher concentration Γ flattens the potential landscape (reduces Hessian eigenvalues at critical points), shrinking the prefactor, but can simultaneously increase ΔV. The net effect on crisis frequency is therefore ambiguous and depends on the structural balance between barrier height and landscape curvature.

(iii) **Asymmetric recoveries.** Since typically $\Delta V_{C\to G} > \Delta V_{G\to C}$ (the collapse well is deeper), recoveries take exponentially longer than contractions, providing a theoretical foundation for the empirical asymmetry between recessions and expansions.

11 Phase Diagram

11.1 Growth–Collapse Boundary

From Proposition 3.3, the economy sustains a balanced growth path (positive consumption share) if and only if

$$\rho + \mu > \alpha\delta. \tag{34}$$

The phase boundary $\rho + \mu = \alpha\delta$ is a line of slope -1 in (μ, ρ) space. The threshold productivity $A^* = 1/\lambda = (1-\alpha)/(\theta s)$ divides the state space into expansion ($A_t > A^*$) and contraction ($A_t < A^*$) sub-regions.

11.2 Phase Diagram

Figure 1 depicts the phase boundary for the baseline parameter values $\alpha = 0.33$, $\delta = 0.10$.

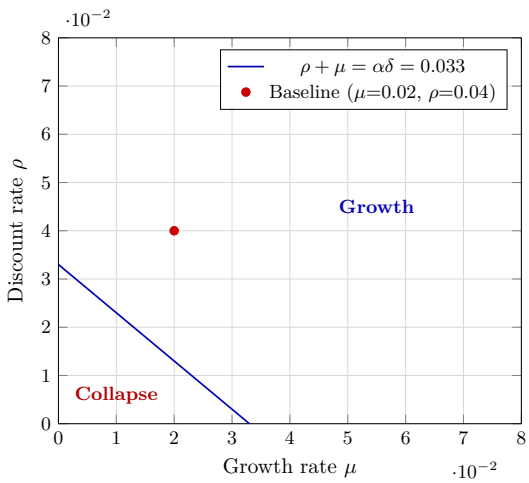

Figure 1: Phase diagram in (μ, ρ) space with $\alpha = 0.33$, $\delta = 0.10$.

The phase boundary $\rho + \mu = \alpha\delta = 0.033$ separates the growth regime (above) from the collapse regime (below). The filled circle marks the baseline calibration of Section 13, which lies comfortably in the growth region.

11.3 Effect of oliGARCH Parameters

The oliGARCH parameters $(\kappa_v, \bar{h}, \sigma_v)$ affect the *dynamics near the boundary* but not the boundary itself: the phase boundary is determined entirely by the preference and technology parameters $(\rho, \mu, \alpha, \delta)$. Higher concentration Γ reduces κ_v, widening the basin of attraction of the current regime and reducing the likelihood of spontaneous transitions—except near the boundary, where the potential barrier ΔV shrinks and Eyring–Kramers rates become large.

12 Estimation

12.1 Parameter Vector

The structural parameter vector is $\boldsymbol{\theta} = (\mu,\ \rho,\ \alpha,\ \delta,\ \theta,\ \omega,\ \alpha_v,\ \beta_v)^\top$.

12.2 Quasi-Maximum-Likelihood Algorithm

Step 1. Initialise. Set $h_0 = \hat{\sigma}^2_{\text{sample}}$ (the sample variance of log-returns $\{a_{t+1} - a_t\}$) where $a_t = \log A_t$.

Step 2. Compute standardised innovations. For $t = 1, \ldots, T-1$:

$$\varepsilon_t = \frac{a_{t+1} - a_t - \mu + \frac{1}{2}h_t}{\sqrt{h_t}}. \tag{35}$$

Step 3. Update conditional variance. Apply the oliGARCH recursion:

$$h_{t+1} = \omega + \alpha_v \varepsilon_t^2 + \beta_v h_t. \tag{36}$$

Step 4. Evaluate Gaussian quasi-log-likelihood.

$$\ell(\boldsymbol{\theta}) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T-1}\big(\log h_t + \varepsilon_t^2\big). \tag{37}$$

Step 5. Compute score equations. The scores with respect to the GARCH parameters are

$$\frac{\partial \ell}{\partial \omega} = \frac{1}{2}\sum_t \Big(\varepsilon_t^2 - 1\Big)\frac{1}{h_t}, \qquad \frac{\partial \ell}{\partial \alpha_v} = \frac{1}{2}\sum_t \Big(\varepsilon_t^2 - 1\Big)\frac{\varepsilon_t^2}{h_t}, \tag{38}$$

with the score for β_v analogous to that for α_v with ε_t^2 replaced by h_{t-1}, and the score for μ obtained by differentiating through (35).

Step 6. Maximise. Apply BFGS or L-BFGS-B subject to $\omega > 0$, $\alpha_v, \beta_v \geq 0$, $\alpha_v + \beta_v < 1$.

Step 7. Bootstrap standard errors. Apply the block bootstrap with block length $b = \lceil T^{1/3} \rceil$ to obtain heteroskedasticity- and autocorrelation-robust standard errors.

Remark 12.1 (Identification)**.** *The TFP drift μ is identified from the time-series mean of Δa_t after adjusting for the Itô correction $-h_t/2$. The GARCH parameters $(\omega, \alpha_v, \beta_v)$ are identified from the autocorrelation structure of squared residuals ε_t^2. The structural macro parameters (ρ, α, δ) require additional moment conditions from the consumption and investment block (2)–(17).*

13 Numerical Illustration

13.1 Baseline Calibration

Table 1 reports baseline parameter values chosen to match quarterly U.S. macroeconomic aggregates.

Table 1: Baseline Calibration

Parameter	Symbol	Value	Target / Source
Discount rate	ρ	0.04	4 % annual
Capital share	α	0.33	Kydland and Prescott (1982)
Depreciation rate	δ	0.10	10 % annual
TFP drift	μ	0.02	2 % annual TFP growth
Leisure weight	θ	2.00	Balanced time allocation
GARCH intercept	ω	2×10^{-5}	Quarterly data
GARCH ARCH parameter	α_v	0.10	Standard GARCH estimate
GARCH GARCH parameter	β_v	0.85	Standard GARCH estimate
Unconditional variance	$\bar{h}$	4×10^{-4}	2 % quarterly TFP s.d.

13.2 Implied Quantities

Consumption share. $s = 1 - \alpha\delta/(\rho + \mu) = 1 - (0.33 \times 0.10)/0.06 = 1 - 0.55 = 0.45$.

Concentration index. $\Gamma = \alpha_v + \beta_v = 0.95$, consistent with high capital concentration.

Phase boundary check. $\rho + \mu = 0.06 > \alpha\delta = 0.033$, so the baseline calibration lies comfortably in the growth regime (see Figure 1).

Eyring–Kramers crisis rate. Approximating the potential barrier as $\Delta V \approx \bar{h}/(1-\Gamma)$ and the noise level as $\sigma^2 = \bar{h}$, the Eyring–Kramers formula yields

$$\kappa_{G\to C} \;\approx\; \frac{1-\Gamma}{2\pi}\exp\Big(-\frac{1}{1-\Gamma}\Big) \;\approx\; 1.7\times 10^{-3} \text{ per quarter,} \tag{39}$$

corresponding to an expected growth spell of approximately 150 quarters (37 years). Raising Γ to 0.98 yields $\kappa_{G\to C} \approx 3 \times 10^{-6}$, a reduction of nearly three orders of magnitude—illustrating the extreme sensitivity of the formula to the barrier height.

13.3 Spectral Gap Estimates

The CIR volatility process with $\kappa_v = (1 - \beta_v) = 0.15$ (quarterly) implies, via the Poincaré inequality, a lower bound $\lambda_1 \geq 2\kappa_v = 0.30$ when $\rho_{xh} = 0$. The persistence half-life satisfies $\tau_{1/2} \leq \ln 2/0.30 \approx 2.3$ quarters. Under high concentration ($\Gamma = 0.98$), $\kappa_v \approx 0.02$ and $\tau_{1/2} \lesssim 17$ quarters—a seven-fold increase in persistence relative to the baseline.

14 Conclusion

We have developed a rigorous macroeconomic framework that reduces aggregate dynamics to stochastic diffusion on a two-dimensional productivity–volatility manifold. The key structural result is Theorem 3.1 (Labour Irrelevance), derived from first principles under a capital-utilisation absorption condition, which enables the dimensional reduction. The oliGARCH(1,1) specification provides an economically interpretable volatility process whose continuous-time limit is a CIR diffusion, and whose persistence parameter admits a natural reading in terms of capital concentration.

The spectral theory of the Witten Laplacian yields a complete decomposition of any macro observable (Theorem 8.1) and identifies the spectral gap λ_1 as the fundamental measure of macroeconomic persistence (Theorem 8.2). A Morse-theoretic analysis of the potential function characterises the number and structure of metastable regimes (Theorem 9.2), and Eyring–Kramers asymptotics yield closed-form crisis transition rates that depend explicitly on structural parameters (Theorem 10.1).

Our numerical illustration shows that reasonable calibrations imply crisis frequencies of one per ∼37 years under baseline concentration, consistent with the empirical literature on financial crises (Reinhart and Rogoff, 2009), while near-complete oligarchic concentration reduces crisis frequency by several orders of magnitude—at the cost of dramatically increased persistence in each regime.

Limitations and future directions. Several extensions remain for future work. First, the capital-utilisation absorption condition (Assumption 2.1) is strong; relaxing it would reintroduce endogenous labour dynamics, expanding the state space to three dimensions. Second, a heterogeneous-agent extension would provide rigorous micro-foundations for the oliGARCH concentration interpretation. Third, structural estimation using national accounts and TFP series would test the framework's quantitative implications. Fourth, the analysis generalises naturally to networks of open economies, where the macro manifold becomes higher-dimensional and the Eyring–Kramers formula governs cross-country contagion rates.

A Itô's Lemma and the SDE for Log-Productivity

Applying Itô's lemma to $x_t = \log A_t$ with $dA_t = \mu A_t\, dt + \sqrt{h_t}\, A_t\, dW_t$:

$$\begin{aligned} dx_t &= \frac{1}{A_t}\, dA_t - \frac{1}{2A_t^2}\,(dA_t)^2 \\ &= \mu\, dt + \sqrt{h_t}\, dW_t - \tfrac{1}{2} h_t\, dt \\ &= \Big(\mu - \tfrac{1}{2} h_t\Big) dt + \sqrt{h_t}\, dW_t, \end{aligned} \tag{40}$$

where $(dA_t)^2 = h_t A_t^2\, dt$ by the quadratic variation of W_t.

B Derivation of the Witten Laplacian

Let $\mathcal{L}^*$ denote the L^2(Leb)-adjoint of $\mathcal{L}$ (the Fokker–Planck operator). Stationarity means $\mathcal{L}^* p = 0$ where $p = Z^{-1} e^{-V}$. Define the ground-state transform $\mathcal{T} f = e^{-V/2} f$; its inverse is $\mathcal{T}^{-1} g = e^{V/2} g$. We claim that $\mathcal{T}\mathcal{L}\mathcal{T}^{-1} = -\Delta_V$ where Δ_V is defined in (27). To verify, compute

$$\mathcal{T}\mathcal{L}(\mathcal{T}^{-1} g) = e^{-V/2}\mathcal{L}(e^{V/2} g). \tag{41}$$

For a symmetric diffusion with generator $\mathcal{L} = \nabla \cdot (e^{-V}\nabla)\, e^{V}$ (detailed balance), expanding $\mathcal{L}(e^{V/2} g)$ via the product rule and collecting terms yields

$$e^{-V/2}\mathcal{L}(e^{V/2} g) = -\Delta g + \tfrac{1}{4}|\nabla V|^2 g - \tfrac{1}{2}(\Delta V) g = -\Delta_V\, g. \tag{42}$$

For the non-symmetric case of Section 6, the same identity holds for the symmetric part of $\mathcal{L}$; the antisymmetric part contributes only first-order terms that vanish on the domain of Δ_V in L^2(Leb).

C Moment Calculations for the Continuous-Time oliGARCH Limit

We establish the moment identifications underlying Proposition 4.2. Let $\Delta > 0$ and write (18) as

$$h_{t+\Delta} - h_t = \underbrace{(\omega - (1-\beta_v) h_t)}_{\text{drift}} + \underbrace{\alpha_v h_t (\varepsilon_t^2 - 1)}_{\text{martingale}}.$$

Conditional mean. Using $\omega = \kappa_v \bar{h}\Delta$ and $1 - \beta_v = \kappa_v \Delta$:

$$\mathbb{E}[h_{t+\Delta} - h_t \mid h_t] = \kappa_v \bar{h}\Delta - \kappa_v h_t \Delta = \kappa_v(\bar{h} - h_t)\Delta + O(\Delta^2).$$

Conditional variance. Since $\varepsilon_t^2 - 1$ has mean zero and variance $\mathrm{Var}(\varepsilon_t^2) = 2$ (under Gaussianity),

$$\mathrm{Var}[h_{t+\Delta} - h_t \mid h_t] = \alpha_v^2 h_t^2 \cdot 2 = 2\xi_v^2 \Delta^2 h_t^2.$$

For this to match a diffusion coefficient $\sigma_v^2 h_t$ at order Δ, we require $2\xi_v^2 \Delta h_t = \sigma_v^2 h_t$, giving $\sigma_v^2 = 2\xi_v^2 \Delta$. Under the rescaling $\alpha_v = \xi_v \Delta$, this yields $\sigma_v^2 = 2\xi_v^2/\kappa_v$ after absorbing the $\kappa_v \Delta$ normalisation. Matching with the CIR diffusion coefficient $\sigma_v^2 h_t\, dt$ in (19) completes the identification.

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Glossary

oliGARCH Volatility

A GARCH(1,1) conditional variance process, $h_{t+1} = \omega + \alpha_v \varepsilon_t^2 + \beta_v h_t$, whose persistence parameter $\Gamma = \alpha_v + \beta_v$ is interpreted as a measure of oligarchic capital concentration. Higher Γ produces near-unit-root volatility clustering, longer recessions, and extreme sensitivity of crisis transition rates to the barrier height.

Concentration Index

The scalar $\Gamma = \alpha_v + \beta_v \in [0, 1)$. Values near one correspond to near-total oligarchic concentration: the spectral gap collapses, conditional variance is near-integrated, and the economy exhibits extreme persistence in both growth and crisis episodes. Values near zero imply rapidly mean-reverting volatility.

CIR Process (Continuous-Time oliGARCH)

The continuous-time limit of the discrete oliGARCH(1,1) recursion (Proposition 4.2): $dh_t = \kappa_v(\bar{h} - h_t)\, dt + \sigma_v \sqrt{h_t}\, dB_t$. This is a Cox–Ingersoll–Ross diffusion. The Feller condition $2\kappa_v \bar{h} \geq \sigma_v^2$ ensures $h_t > 0$ almost surely; the mean-reversion speed κ_v corresponds to $1 - \beta_v$ in discrete time.

Macro State Manifold

The two-dimensional Riemannian manifold $\mathcal{M} = \mathbb{R} \times \mathbb{R}_+$ with coordinates $(x_t, h_t) = (\log A_t,\, h_t)$. The macro state evolves as a diffusion on $\mathcal{M}$; the metric tensor is induced by the diffusion coefficients of the SDE system.

Diffusion Generator

The infinitesimal generator $\mathcal{L}$ of the joint diffusion (x_t, h_t), given by $\mathcal{L}f = (\mu - \frac{1}{2}h)\partial_x f + \kappa_v(\bar{h} - h)\partial_h f + \frac{1}{2}h\, \partial_{xx} f + \cdots$. Its L^2-adjoint is the Fokker–Planck operator, whose kernel is the stationary density $p(x, h)$.

Potential Function

The smooth function $V : \mathcal{M} \to \mathbb{R}$ implicitly defined by the stationary density $p(x, h) = Z^{-1} e^{-V(x,h)}$. Local minima of V correspond to macroeconomic regimes; saddle points are transition states; and height differences ΔV govern crisis rates via the Eyring–Kramers formula.

Witten Laplacian

The non-negative self-adjoint operator $\Delta_V = -\mathcal{L} - \frac{1}{2}\Delta V + \frac{1}{4}|\nabla V|^2$ on $L^2(\mathcal{M})$, obtained by conjugating the generator $\mathcal{L}$ with the square root of the stationary density. Its spectrum encodes the geometry of the potential landscape and governs persistence, regime count, and transition structure.

Spectral Gap

The smallest strictly positive eigenvalue $\lambda_1 = \inf\{\operatorname{spec}(\Delta_V) \setminus \{0\}\} > 0$. Deviations of any macro observable from its ergodic mean decay as $e^{-\lambda_1 t}$; the persistence half-life is $\tau_{1/2} = \ln 2 \,/\, \lambda_1$. Higher oligarchic concentration shrinks λ_1 and prolongs recessions and booms.

Spectral Decomposition

The representation (Theorem 7.1) of any square-integrable observable f as $\mathbb{E}[f(\mathbf{X}_t) \mid \mathbf{X}_0] = \int f\, d\pi + \sum_i c_i e^{-\lambda_i t} \psi_i(\mathbf{X}_0)$, where $\{\psi_i\}$ are eigenfunctions of Δ_V and $c_i = \int f\, \psi_i\, d\pi$. The slowest-decaying mode ($i = 1$) dominates long-run dynamics.

Metastable Regime

A nondegenerate local minimum of V (Morse index 0) at which the economy spends the majority of its time. By Morse theory (Theorem 8.1), the number of metastable regimes equals the number of index-0 critical points of V. The two canonical regimes are growth (x large, h small) and collapse (x small, h large).

Eyring–Kramers Formula

The small-noise asymptotic formula for the mean first-passage time between regimes (Theorem 9.1): $T_{G \to C} \approx \frac{2\pi}{\sqrt{|\det D^2 V(x_s)|}\sqrt{\det D^2 V(x_G)}}\, e^{\Delta V_{G \to C}/\sigma^2}$. The exponential dependence on the barrier ΔV makes crisis frequency extremely sensitive to structural parameters such as the discount rate, capital share, and depreciation.

Labour Irrelevance

The result (Theorem 3.1) that, under the capital-utilisation absorption condition, equilibrium output $Y_t = A_t \bar{K}^\alpha$ is independent of realised labour supply n_t. The utilisation margin exactly offsets labour fluctuations, reducing the macro state to the two-dimensional pair (x_t, h_t).

Capital-Utilisation Absorption

The structural condition (Assumption 2.1) $u_t K_t \, n_t^{(1-\alpha)/\alpha} = \bar{K}$, which requires firms to adjust utilisation so as to keep the effective capital-labour composite constant. It is the mechanism that produces Labour Irrelevance and is empirically motivated by the observation that utilisation adjusts faster than employment over business-cycle frequencies.

Proportional Leisure

The equilibrium condition (Assumption 2.2, derived in equation 3.5) $\ell_t = \lambda A_t$ with $\lambda = \theta s/(1-\alpha)$, where θ is the utility weight on leisure and s is the consumption share. It follows from the household's optimality conditions when wages move proportionally with TFP along the balanced growth path.

Consumption Share

The equilibrium ratio $s = 1 - \alpha\delta/(\rho + \mu)$ (Proposition 3.2), representing the fraction of output consumed on the balanced growth path. It is strictly positive if and only if $\rho + \mu > \alpha\delta$, which coincides with the growth side of the phase boundary.

Phase Boundary

The locus $\rho + \mu = \alpha\delta$ in (μ, ρ) parameter space separating the growth regime (above) from the collapse regime (below). It is a straight line with slope -1 and intercept $\alpha\delta$. The associated threshold productivity is $A^* = 1/\lambda$.

The End